

Vinod Kumar Gorle

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SUMMARY

AI / GenAI Engineer with nearly 4 years of experience building AI applications using LLMs, VLMs, RAG, multimodal AI, semantic search, and intelligent retrieval. Experienced with LangChain, Azure AI Foundry, Azure AI Search, Azure Machine Learning, FastAPI, LLaMA-3, Mistral, Qwen-VL, LLaVA, BGE-M3, SigLIP, FAISS, and Sentence Transformers. Skilled in AI architecture, query understanding, hybrid retrieval, embeddings, VLM-based image analysis, prompt engineering, model evaluation, retrieval optimization, secure deployment, and enterprise AI solutions.

SKILLS

GenAI, LLMs & RAG:

Generative AI, Agentic AI, LLMs, VLMs, RAG, Multimodal AI, AI Architecture, LangChain, LangGraph, Prompt Engineering, Prompt Injecting, Query Understanding, Intent Classification, RAG Evaluation

Search & Retrieval:

Semantic Search, Lexical Search, Hybrid Search, Vector Search, Multimodal Retrieval, Embeddings, Reranking, Query Expansion, Retrieval Optimization, FAISS, Azure AI Search, Snowflake Cortex Search, Snowflake Cortex Analyst, Snowflake Arctic Embed L v2.0, Voyage Multimodal 3, Snowflake Functions all-MiniLM-L6-v2, Sentence Transformers, Text-to-SQL, Vector Database, Voice-To-Text

AI/ML & Computer Vision:

Machine Learning, Deep Learning, PyTorch, Scikit-learn, NLP, Computer Vision, YOLOv5, OpenCV, OpenVINO, Object Detection, Object Tracking, Video Analytics, Edge AI, Model Optimization, FP16, TensorFlow, Hugging Face, PyTorch, Data pre-processing, Feature Engineering, Advanced Data Analysis.

Azure AI & Cloud:

Azure AI Foundry, Azure Machine Learning, Azure AI Search, Azure Data Factory Azure Blob Storage, Microsoft Entra ID, Azure OpenAI, Gemini models, Managed Endpoints, AI Model Deployment, Model Lifecycle Management

Cloud & DevOps:

Microsoft Single Sign-On (SSO), Microsoft Azure, Azure App Services, Azure Functions, Azure Storage, Azure Compute, Cloud AI Architecture, Cloud Deployment, Docker, Linux, Git, GitHub, CI/CD, API Deployment, AI/ML Model Serving, Scalable AI Inference

Backend & Data:

Python, FastAPI, Flask, REST APIs, API Architecture, Asynchronous Processing, MS-SQL, MySQL, Database Design, Query Optimization, Data Pipelines, Dagster Automation

AI Development:

Cursor, OpenAI Codex, GitHub Copilot, AI Coding Agents, AI-Assisted Development, Code Generation, Code Refactoring

Full-Stack:

React Native, React JS, Node JS, JavaScript, HTML, CSS, jQuery, Bootstrap, WordPress

EXPERIENCE

SOFTWARE ENGINEER — Xinthe Technologies (June 2023 – Present)

❖ Enterprise AI Logo Search & Multimodal Retrieval Platform - Team Lead

Python | FastAPI | Azure | Snowflake | Snowflake Cortex AI | Azure Data Factory | Azure Blob Storage | Azure AI Search | LLMs | VLMs | Snowflake Arctic Embed L v2.0 | Voyage Multimodal 3 | FAISS | SQL Server | Microsoft Entra ID

- Led and architected an enterprise AI-powered Logo Search platform for intelligent retrieval of logos and designs using multimodal AI, semantic search, hybrid retrieval, metadata filtering, and image similarity.
- Designed the end-to-end data pipeline using Azure Data Factory for automated file/data synchronization, Azure Blob Storage for centralized image storage, and Snowflake for structured data management and processing.
- Implemented VLM-based image understanding and metadata generation, using Snowflake Arctic Embed L v2.0 for text and caption embeddings and Voyage Multimodal 3 for image embeddings, enabling semantic and multimodal retrieval.
- Built the search and retrieval pipeline using Snowflake Cortex AI, Azure AI Search, FAISS, Python, and FastAPI, supporting query understanding, semantic retrieval, lexical search, metadata filtering, query expansion, ranking, and reranking.
- Evaluated LLM/VLM-based approaches for intent understanding, image captioning, semantic retrieval, and query-to-SQL, validating accuracy, performance, scalability, and cost through POCs.
- Implemented Microsoft Entra ID, SSO, and RBAC for secure enterprise authentication and role-based access.
- Led the technical team by defining architecture, coordinating development activities, assigning tasks, reviewing implementations, conducting technical research, and driving the project through development, POC, testing, and final implementation

❖ Enterprise Document Q&A & RAG Platform

FastAPI | LLaMA-3 | Mistral | Ollama | LangChain | RAG | all-MiniLM-L6-v2 | Sentence Transformers | FAISS | Python | SQL | Azure | Embeddings

- Developed a self-hosted GenAI Document Q&A platform using LLaMA-3, Mistral, Ollama, LangChain, FastAPI, and Python, supporting offline LLM inference, RAG, and knowledge-grounded question answering.
- Designed an end-to-end RAG pipeline covering document ingestion, parsing, preprocessing, chunking, embeddings, FAISS vector indexing, Top-K semantic retrieval, query understanding, context construction, prompt engineering, and LLM response generation.
- Optimized retrieval and response quality using all-MiniLM-L6-v2, similarity thresholds, chunk size/overlap, Top-K/Top-P, reranking, and retrieval parameters; evaluated groundedness, faithfulness, relevance, correctness, accuracy, hallucination, latency, and inference performance.

❖ Python Backend & AI/ML Application Architecture

Python | FastAPI | Flask | Django | PyTorch | Scikit-learn | Machine Learning | Deep Learning | NLP | Computer Vision | Transformers | Model Training | Model Inference | Model Evaluation | REST APIs | MS-SQL | Async Processing

- Designed scalable Python backend and AI/ML architectures using FastAPI, Flask, Django, PyTorch, and Scikit-learn for data processing, model training, evaluation, inference, NLP, Computer Vision, and LLM/VLM services.
- Built production-ready enterprise services with MS-SQL, REST APIs, asynchronous workflows, authentication, model deployment, CI/CD, vector search, embeddings, and GenAI integrations, focusing on scalability, performance, and reliability.

❖ Computer Vision & Edge AI Architecture

YOLOv5 | OpenCV | OpenVINO | Object Detection | Tracking | Edge AI

- Designed real-time computer vision and video analytics pipelines for object detection, people counting, surveillance, tracking, ROI-based analysis, and event detection, achieving 90%+ counting accuracy through optimized tracking and post-processing.
- Architected edge AI inference using Intel OpenVINO with FP16/INT8 optimization for low-latency, high-FPS deployment, integrating vision inference with backend APIs and real-time analytics while optimizing latency, throughput, and hardware utilization.

❖ Full-Stack & Web Application Engineering

React | JavaScript | PHP | WordPress | HTML | CSS | jQuery

- Designed modular full-stack architectures integrating React/JavaScript frontends with Python backend services, REST APIs, enterprise data workflows, and third-party integrations.
- Developed reusable React/JavaScript components and responsive interfaces using jQuery, HTML, CSS, and Bootstrap for interactive enterprise dashboards and business applications.
- Built and customized WordPress applications with reusable UI components and API integrations, working across frontend, backend, API, and database layers for end-to-end application development.

SOFTWARE ENGINEER INTERN — Xinthe Technologies (Nov 2022 – May 2023)

- Performed QA testing, model validation, inference benchmarking, and performance evaluation for Automatic License Plate Recognition (ALPR) systems using live and recorded video streams, identifying detection and inference issues to improve model reliability.
- Designed and developed real-time people counting and video analytics pipelines using Intel OpenVINO, OpenCV, and computer vision techniques for surveillance and edge AI applications.
- Created and maintained annotated datasets, performed model validation, error analysis, inference output verification, and performance evaluation to improve computer vision model accuracy and deployment readiness.

CERTIFICATIONS & TRAINING

- Develop Generative AI Solutions with Azure OpenAI Service – Microsoft Learn
- Develop Generative AI Apps in Azure – Microsoft Learn
- Deploy and Manage Machine Learning Models with Azure Machine Learning – Microsoft Learn
- Explore Azure AI Foundry – Microsoft Learn
- C and Python Programming – Cisco

WEBSITES & SOCIAL LINKS

- Portfolio: <https://portfolio-vinodkumargorle.vercel.app/>
- LinkedIn: <https://www.linkedin.com/in/vinodkumargorle8367/>
- GitHub: <https://github.com/vinodkumargorle/>

EDUCATION

- **MCA – Master of Computer Applications (2020–2022)**
Gayatri Vidya Parishad College of Engineering, Visakhapatnam, GPA: **7.6 / 10**
- **B.Sc. – Computer Science (2017–2020)**
Sri GCSR College, Rajam, Andhra Pradesh, GPA: **7.8 / 10**

